Assignment 2.

1. Case Study: P&L impacts for an IRS
Bootstrap the IR curve on the 15th of February 2008 at 10:45 C.E.T. up to the last swap rate available, using Depos, STIR futures and Swap rates. Outputs: dates, discounts, zeroRates. Include in the "dates" only end dates of underlying contracts.

Bank XX uses a nonstandard methodology in order to obtain discount factor curve. Let us consider market situation on the 15th of Feb 2008 at 10:45 a.m. CET. Let us suppose that using bank discount factor curve one obtains a 6y swap rate vs 3M equal to 4.027% (versus a mid-market rate of 4.127%).

Q1: Can we estimate the P&L for a trader that receives a 6y vs 3M at 4.117% (mkt conventions) for 100 Mln € at 10:45 of the 15th of Feb 2008? Can we say something about the quality of this estimation?
Q2: Bootstrap is not the only technique to obtain Discount Factors (DFs) from quoted rates. Why is it so relevant the bootstrap of DFs in finance? Why is it so important to obtain a discount factor curve that reproduces exactly liquid instruments?

2. Exercise: "I.B." Coupon Bond
Price an 8y "I.B. coupon bond" issued on the 31st of Jan '23 with coupon rate equal to the corresponding mid-market 8y swap rate. Assume for the coupons a 30/360 European day count and a Face Value 100 Mln; it has same expiry, fixed rate & reset dates of the IRS.
Hint: Shortcuts are appreciated.

3. Exercise: Asset Swap
Given the discounting curve vs Euribor 3m on the 15th of February 2008 at 10:45 C.E.T. and knowing that knowing that the clean price of a bond with maturity 31st of March 2012 for an issuer YY is 101.5 (101.5% of the *face value*) with an annual coupon equal to 4.6% (*annual bond unadjusted modified following*), compute the *Asset Swap Spread Over Euribor3m*. The bond has been issued on the 31st of March 2007.

4. Case Study: CDS Bootstrap
Given values for discounts on the case-study on curve bootstrap the 15th of February 2008 at 10:45 C.E.T. consider the obligor ISP with a recovery π equal to 40% and CDS spreads (annual bond): 1y 29 bps, 2y 34 bps, 3y 37 bps, 4y 39 bps, 5y 40 bps, 7y 40 bps.

a. Build a complete set via a spline.
b. Build $\lambda(t)$ piecewise constant for the issuer, neglecting the "accrual" term.
c. Which is the impact of the "accrual" term? Show that this term is really negligible.
d. Consider Jarrow-Turnbull approximation (a constant λ and continuously paid CDS spread) and compare the result with the one previously obtained.

5. Exercise: Credit Simulation
Given *survival probability* at time t up to T (30 years): $P(t,T) = e^{-\int_t^T \lambda_s \, ds}$;
with $\lambda_s = \lambda_1 \, 1_{s \le \theta} + \lambda_2 \, 1_{s > \theta}$ and $\lambda_1 = 4$ bps, $\lambda_2 = 10$ bps, $\theta = 5 \, years$.

a. Simulate the *default time* τ.
b. Fit the survival probability.

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Fit the distribution of default times using a sample of $M=10^5$ points. Provide Estimator and Confidence Interval (CI) on the Estimator (only for λ_1 and λ_2, not for θ).
Hint: A possibility (not binding) is to consider a plot of the "experimental" survival probability and the one obtained from the fit (in a loglinear scale) with a CI. Another one is to consider the ML estimator and its CI.

6. Exercise
With the discounting curve you've obtained at point 1, compute the NPV of a set of monthly cash flows received on the 19th of each month for 20y, starting from March 2008, with an Average Annual Growth Rate (AAGR) of 5%, in two cases:

a. when the initial monthly cash flow is 1.5K €;
b. when the initial monthly cash flow is 6.0K €.

The AAGR growth rate is applied each year in March.

7. Arbitrage
Paper [1] presents a strategy of a synthetic Repo (Collateralized load) that should lead to an arbitrage. Can you describe it and explain why the strategy leads to arbitrage? Why does not the arbitrage disappear? [Use 6 lines MAX]

[1] Azzone, M., Baviera, R., & Manzoni, P. (2025). The puzzle of Carbon Allowance spread. Energy Economics, 146, 108459. Available on-line https://doi.org/10.1016/j.eneco.2025.108459

Function signatures
[dates, discounts, zeroRates] = bootstrap(datesSet, ratesSet);
[datesCDS, survProbs, intensities] =
bootstrapCDS(datesDF, discounts, datesCDS, spreadsCDS, flag, recovery).

dates and discounts are the same outputs of bootstrap function; datesCDS do not include the settlement date; function outputs are vectors with the same length; flag = 1 (approx), 2 (exact) or 3 (JT).

All vectors are column vectors.

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